

Third Permutation

Time limit: 1.00 s

Memory limit: 512 MB

You are given two permutations a and b such that $a_i \neq b_i$ in every position. Create a third permutation c such that $a_i \neq c_i$ and $b_i \neq c_i$ in every position.

Input

The first line has an integer n : the permutation size.

The second line has n integers $a_1, a_2, \dots, a_n$.

The third line has n integers $b_1, b_2, \dots, b_n$.

Output

Print n integers $c_1, c_2, \dots, c_n$. You can print any valid solution. If there are no solutions, print IMPOSSIBLE.

Constraints

- $2 \leq n \leq 10^5$

Example

Input	Output
5 1 3 2 5 4 4 1 3 2 5	3 2 5 4 1